

TSSSU Final Report (Reporting period: 1 April 2024 – 30 March 2025)
Submitted by (TadReamk Limited) on (31 May 2025)

*Please complete this report AND provide relevant supporting documents for justification. Thank you.

1. Executive summary of the progress

At this point in the operation, the start-up has experienced exponential growth and made remarkable progress in every category, both in terms of technology and business development.

In terms of Human Resources, the company now has a stable workforce and has further strengthened its core team with minimal personnel changes. This demonstrates our ability not only to attract top talent but also to retain them.

In terms of AI product development, we have made significant advancements across the board, including enhancements to TadReamk Eye (TE), TadReamk Create (TC).

We have built various supporting software. For example, we have developed our proprietary logo labelling tool, which currently serves as supporting software for TE. As a spin-off project, we aim to eventually market it as a standalone product. This tool was previously referred to as the 1 Million Logo Database (IMLD) initiative in the previous final report.

Another notable additional developments during this phase is the introduction of the MallBuddy chatbot. This innovation stemmed from our exceptional logo development capabilities for TE, naturally extending to shopping mall logo recognition and logo-based navigation. This expansion is a natural progression and aligns well with our main products. Additionally, it has the potential to generate additional revenue streams and open up numerous new opportunities for us.

While steadfastly adhering to our original goals as outlined by the TSSSU objectives, our company has significantly realigned its priorities in response to two major business opportunities:

1. Opportunity 1:

At the end of 2024, during the HK Chief Executive's policy address, which states that,

"89. Next year, the Trade Marks Registry under the Intellectual Property Department (IPD) will launch a new AI-assisted image search service to facilitate the public's search of the trademark database."

This announcement was both exciting and unexpected. As Hong Kong's leading AI trademark startup, we must seize this golden opportunity and have

therefore re-prioritized to be fully prepared for a potential tender. We have set aside other logo-based technologies to focus exclusively on making TE ultra-capable and highly efficient in terms of search. Our goal is to achieve a level of excellence that significantly surpasses what was initially necessary for the Patent Attorney market.

2. Opportunity 2:

By the end of 2024, we shifted our priority from targeting individual patent attorneys to a more lucrative market: cross-border e-commerce companies. Many of these companies are based in Shenzhen and aim to sell Chinese products overseas, such as in Amazon, where trademark infringement is a major concern. Since then, we have visited numerous cross-border e-commerce companies in Shenzhen, all of whom have praised our company's technologies.

Therefore, with the emergence of new opportunities and our swift strategic responses, the latter half of our business plans has been significantly re-prioritized, resulting in a clearer focus on three distinct target markets:

i. For the Government (G) market:

We are preparing to pitch to the Hong Kong Intellectual Property Department (HK IPD) and demonstrate the robust capabilities of our Trademark Eye, a multi-perspective image-based trademark similarity search tool. Additionally, we are exploring potential partnerships to co-pitch to the HKIPD.

ii. For the Business (B) market:

Our efforts are concentrated on the cross-border e-commerce sector, with tailored product offerings designed to better serve its specific needs.

iii. For the Consumer (C) market:

Strictly speaking, our customer base doesn't include individual consumers, as those seeking trademark registration are primarily businesses. However, we refer to this sector as the to-C market for differentiation. This is our initially targeted market sector, comprised of individual businesses seeking to register their trademarks and the patent attorneys who represent them.

For the remainder of this document, we will detail our progresses in various aspects.

2. Deliverables/technological and financial achievements

In terms of software development:

TadReamk Eye (TE):

TE is a tool designed for automated trademark similarity searches. Initially, our target audience is the consumer market, specifically individual businesses seeking to register trademarks and/or patent attorneys representing them.

With this goal in mind, our development objective until the end of 2024 was to launch the product quickly for revenue generation, even if it wasn't revolutionary or extremely sophisticated, as the to-business sector is a highly subjective market: We therefore have completed significantly enhanced versions of both the mobile and web apps, offering improved functionality. The mobile app is available for both iOS and Android platforms. After receiving feedback from potential customers during the alpha-testing phase, particularly patent attorneys, our TE feature now includes many additional capabilities, such as multi-dimensional perspectives for trademark searches. We've also resolved several functional issues to ensure compatibility with modern hardware (e.g., iOS's automated focusing functionality) and addressed non-functional issues to enhance the user interface. We completely revamped our logo detection algorithm by developing our own labeling software and dataset, resulting in significantly improved detection performance for TE.

We successfully launched it to the public in late 2024. We continue to expand the database, which now contains over 30 million entries from the USPTO dataset.

However, with the re-prioritization of TE towards both the business and government sectors (as detailed in the executive summary), particularly, we feel that in order to have any chance to be involved in the Hong Kong IPD tendering process, our product must be the best in the world in terms of search comprehensiveness and efficiency. Therefore, additional development efforts for TE must be undertaken in 2025 from two perspectives:

1. Search Comprehensiveness

After consulting with numerous stakeholders and many laborious experiments, we have extensively expanded our search criteria to include a comprehensive range of features such as Color, Layout, Design, Wording, and Style (CLDWS). Our TE system is now the state-of-the-art.

2. Search Efficiency

For our product to have any chance to be adopted by the government and subsequently used by all its citizens, efficiency is a significant concern. Therefore, we have devoted extensive R&D efforts to enhance search speed. Now, the search of 30 million USPTO data can be returned within seconds.

We are pleased to report that our similarity search is now state-of-the-art, both in terms of comprehensiveness and efficiency.

In addition to ensuring technology readiness, we have prepared to present our products to the Hong Kong IPD and secured opportunities for co-pitching alongside established

industry leaders, such as Clarivate. We are also collaborating with other Hong Kong-based consultancy firms with experience in tendering for government projects.

Now, as our goal is to penetrate both the government and business markets (as detailed in the executive summary), the consumer market has become less significant in terms of revenue generation. However, we recognize its value in enhancing our reputation within the intellectual property community, which in turn supports our efforts when pitching to the government during future tender processes. Consequently, an executive decision has been made to offer services in this sector free of charge for a quicker market penetration.

TadReamk Create (TC):

TC is a tool that employs AI to design new company logos while significantly reducing the risk of infringement.

We have successfully completed the research and development of TC on schedule and within scope. We would like to comment that the R&D process for TC proved more challenging than anticipated. With each new iteration of increasingly sophisticated off-the-shelf large language models, we closely monitored their capabilities and promptly integrated them, thereby avoiding the need to reinvent the wheel. Throughout this process, we have consistently updated our software to remain aligned with the latest model releases.

We have thoroughly investigated various approaches to TC, resulting in a highly stable current iteration with a minimal occurrence of hallucinations, such as unwanted image text in logos. Our product is now publicly available and actively collecting user feedback.

We've incorporated numerous innovations, including the use of Reinforcement Learning (RL) to enhance how prompts are generated for trademark logos. Additionally, the TC user interface has been developed to ensure a professional look and feel.

Logo Detection Dataset Collection and Labeling (formerly one million logo database: 1MLD project)

To perform effective logo detection, we need to collect a vast dataset, particularly for logos in a local context, such as those found in a HK shopping center that may not be available on the internet. Collecting and labeling these logos is an expensive process, but we have now developed a systematic approach to handle this.

The system is highly sophisticated; it is equipped with labeling functionalities for both text and images, and it also includes features for computing billing information for the labeling contractors. It has the potential to be marked as a separate product in the future.

MallBuddy Chatbot

This product began when we realized that Tadreamk Eye's logo detection technology has much broader applications than just trademark detection. Consequently, we expanded our product line to include a shopping center logo-based navigation tool. This tool enhances the user experience by enabling shoppers to localize and navigate the shopping center while also addressing their queries. We believe this extension enriches our already diverse product arsenal with minimal deviation from our core focus. It has been demonstrated to several partners including Sino Group Innovation.

In terms of Technology Achievement:

We have successfully registered our first patent. We have received several awards for our products:

1. Top 100 Award at the SME Innovation and Entrepreneurship Global Contest, Hong Kong Chapter
2. Bronze Medal at the 28th National Invention Exhibition

In terms of Business Achievements: TadReamk has successfully passed multiple rounds of HKSTP incubation assessments and continues to receive support from both TSSSU and HKSTP incubation funding.

3. Milestones achieved (*please align the period and milestones with those mentioned in your approved Application form i.e. Annex A)

Period		Milestones	Achieved? (If not, please explain)
From (dd/mm/yyyy)	To (dd/mm/yyyy)		
01/04/2024	01/06/2024	TadReamk Trademark Detection Database	Completed
01/04/2024	01/08/2024	TadReamk Eye (alpha version) is live: ready for earlier user evaluation.	Completed

Period		Milestones	Achieved? (If not, please explain)
From (dd/mm/yyyy)	To (dd/mm/yyyy)		
01/08/2024	01/12/2024	TadReamk Eye makes its first sale.	Completed. Please note that the sale is not in the form of direct income, but rather its intrinsic value. We have strategically decided (end of 2024) to make TE freely accessible to the public to enhance our reputation within the intellectual property community. As outlined in Sections 1 and 2, this reputation-building and fast market penetration strategy will allow us to swiftly capitalize on a unique and timely opportunity arising from Hong Kong's policy address at the end of 2024, allow us to get ready to tender to the Hong Kong Intellectual Property Department for the AI image searching tool this year.
01/12/2024	01/02/2025	TadReamk Create prototype is finished.	Completed
01/02/2025	01/04/2025	TadReamk Create go lives.	Completed

4. Progress of premise rental, equipment rental/purchase and manpower expenditure

We have progressed the following major equipment:

Here, we list the items procured at this year's budget:

- 1x Apple Mini – This machine is currently dedicated to running our prototype and other proof-of-concept projects.
- 2x additional laptops – These are dedicated to software development and the secure storage of the company's confidential data. They were primarily procured to support the expanding workforce, particularly the new interns over the summer, and will also be used for future interns.
- 1x additional powerful laptop – This Apple Mac Pro was acquired to meet the demanding development deadlines.

In terms of human resource, the company is comprised of the following core members as of 31st/Sep/2024:

- Professor Xu, Yi Da – Founder, primarily oversees resources and strategic direction.
- Mrs. Chan, Linda – Chief Operating Officer, chiefly manages operations and human resources.
- Dr. Nguyen, Alan – Chief Research Officer, responsible for leading research efforts.
- Mr. Niu, Shuai – Member of the research team.
- Mr. Sun, Xiao Yuan – Chief Technical Officer, primarily in charge of development.

The following individuals have been long-term part-time employees since April 2024 (excluding temporary interns):

- Mr. Chen Chun Hang – Researcher
- Ms. Carrie Lin – HR assistant
- Mr. Jason Gong – Research intern

Other than the above workforce, we have also recruited three additional summer STEM interns, two short term technical intern as well as one intern under the YSIP program. The company also engages several contractors on an as-needed basis.

Besides the founder, every team member receives salary fully or partially funded from TSSSU, creating a dream team that excels in all aspect of an AI startup.

The following people have left the company since April 2024:

- Dr. Ningxia Wang, whom we unfortunately had to let go due to delays in obtaining RTH approval in May 2024.
- Mr. Sun Cheng – Business Developer, specialized in business development and the conceptual philosophy of logo design. Mr. Sun departed from the company in late

2024 to pursue personal endeavors. His responsibilities have been partially assumed by Mr. Chen Chun Hang, Ms. Carrie Lin, and Mr. Jason Gong.

5. Any potential collaborations/M&A/investors

The founder, Prof. Xu, boasts strong relationships with numerous investors and venture capitalists. At this stage of the company's development, with multiple products emerging, he has begun engaging with potential investors. Additionally, demonstrating his confidence in the company's future as a major force in the AI industry both in Hong Kong and globally, Prof. Xu has personally invested over HK\$300,000 into the business to date.

Prof Xu, a key opinion leader in AI, has harnessed his online status for the company by initiating an online campaign to crowdsource logo images, aiming to amass a collection of one million logos for his database.

Being located at HKSTP and part of the university startup community has favorably positioned the company to connect with new investors. The Institute for Innovation, Translation and Policy Research (ITPR) of Hong Kong Baptist University (HKBU) and the Knowledge Transfer Office (KTO) teams have significantly enhanced the company's exposure to various investors and funding opportunities.

6. Difficulties encountered

In the 2023-2024 final report, we have documented the difficulties encountered, including:

1. Continuous adaptation of new LLMs
2. Incorporating "device code".
3. Explainability of the design rationale:
4. Challenges in Staffing
5. business difficulties: excessive paperwork
6. Unexpected administrative delays with RTH

While the previous four difficulties have been well managed and are under control in the current phase, the last two issues - "*business difficulties: excessive paperwork*" and "*unexpected administrative delays with RTH*"—remain ongoing challenges that we have come to accept as part of running a startup.

We would also like to highlight several issues that have become prominent in this phase of the company which all have impacted our cashflow:

Firstly, we have observed that although the government offers many grants and initiatives, such as the Talent Hub and the STEM internship program, all these programs require us to pay upfront and seek reimbursement later, sometimes many months later. While we highly appreciate these programs, which have been crucial for our progress, the pay-first-and-reimburse-later model nonetheless adds cash flow pressure on us.

The second challenge is the lack of understanding and support from society at large. While we greatly appreciate the support we receive from ITC, the University, and HKSTP, we feel that not every entity in Hong Kong understands the nature of tech startups founded by professors. For instance, our business bank has not granted us a credit card, even when we applied for a very minimal HK\$20,000 credit limit. Although the reason was not disclosed, we believe it is due to the company currently having no income other than government and HKSTP investment.

Our cash flow situation is typical for a tech startup founded by professors. Therefore, we hoped that the bank could view companies like ours differently from a traditional business's cash flow perspective. Operating without a credit card is very inconvenient, forcing us to purchase items using personal credit cards and then seek reimbursement, which adds unnecessary administrative burden. We suggest that commercial banks develop a better understanding of tech startups supported by TSSSU and establish special programs tailored to these companies. This would help support the government's innovation initiatives more effectively.

Thirdly, a minor issue is that while we highly appreciate the government's initiatives designed to prevent double-dipping of government funding, this can nonetheless add a financial burden to our startup. For example, when we are unable to use TSSSU or HKSTP incubation funding to apply for the Patent Application Grant (PAG), it forces the founder to pay out of pocket.

In summary, combined cash flow challenges present difficulties for the business. However, from a glass-half-full perspective, they are "good problems to have." These challenges arise solely because the company is growing rapidly, making cash flow the only major concern. We believe that the commitment and determination of the founder and the rest of the team will enable us to overcome these obstacles, especially by expediting our go-to-market process.

7. Outstanding issues (if any)

Currently, we are **not** facing any outstanding issues. Our team has successfully navigated through dozens (if not one hundred) technical, business, and operational hurdles to date.

As a startup, we recognize that encountering unexpected challenges is an inevitable part of our journey. Nevertheless, with our growing experience and the expertise of

our talented team, we are confident in our ability to address and overcome any new challenges that may arise.

8. Innovation and technology content and commercialisation

This entrepreneurial venture has spurred a multitude of innovations, some of which we've outlined below:

1. We are in the midst of finalizing the first patent application that details our innovative logo labeling process, a key component of our extensive logo data collection and labeling initiative.
2. Our logo detection algorithm has been completely overhauled from the previous phase, surpassing other state-of-the-art methods described in the current research literature.
3. Our team is actively compiling what we aim to be the world's most comprehensive logo database, a project we've affectionately named the One Million Logo Database (1MLD).
4. We're leveraging the founder's Key Opinion Leader (KOL) status to adopt a crowdsourced approach for gathering logos.
5. The launch of MallBuddy, the logo-based navigation chatbot, further increases TadReamk's exposure in the market and society as a whole. Although this project is a slight diversion from our main path, it holds the potential to create multiple streams of revenue without detracting from our primary objective.
6. Our flagship product, TadReamk Eye (TE), incorporates a multidimensional approach to similarity search, making it arguably the most advanced trademark similarity searching tool ever developed. This claim is supported by our relentless efforts to conduct the on-going exhaustive analysis of our competitors.
7. Our second flagship product, TadReamk Create (TC), offers a unique selling point: any logo generated can be immediately verified by TE upon creation. This ensures they are distinct from existing patented logos, thereby increasing their chances of registration. Our unique algorithm for generating logos without unwanted text has proven to be very successful.

8. Indeed, many tools are being developed using a combination of TE and TC. Moving forward, we are creating a suite of products that leverage these tools with minimal R&D effort.

9. Commercial viability of the business

Our startup holds commercial promise as it operates within a relatively unexplored AI niche market. Reflecting on the global landscape, the volume of trademark applications surged to approximately 15.5 million in 2022, tripling the figure from a decade prior in 2012. This spike underscores the critical demand for tools that can streamline the booming trademark application process.

Many practical examples can be identified that highlight the benefits of our product: for example, conducting a trademark search can be a costly and time-intensive endeavor, with fees ranging from hundreds to thousands of dollars. Consequently, there is a significant advantage in developing an AI-powered tool that can expedite the search process for businesses and patent attorneys, offering them substantial time and cost savings.

Moreover, the creation of trademarks by professional graphic artists is not only costly but also inherently subjective. Crafting an AI tool that can provide artistic rationale for logo creation presents a unique and complex challenge.

Beyond its primary function, the logo detection software we've developed opens doors to a variety of intriguing use cases, such as creating cutting-edge tools that assist consumers in enhancing brand recognition. These auxiliary projects are in progress and are expected to yield results as they continue to evolve.

As detailed in Sections 1, 2, and 3, we underwent a market re-segmentation at the end of 2024, prompted by a timely and unique opportunity presented in the government's policy address. This is further enhanced by the discovery of opportunities within the cross-border e-commerce market.

We have now systematically segmented our marketing strategy into three sectors: Government, Business, and Consumer (see executive summary). Each of these market segments holds significant potential for revenue generation, which we have a rigorous plan for.

10. Capability of your technology start-up and your team to undertake the R&D work and manage the company

We have been fortunate to attract exceptional and experienced individuals who are deemed our most precious assets. Their skills lay a robust groundwork for the company's success, even as we consider exploring additional AI products in the future. As the core members of the team has only changed minimally from the 2023-2024 finally report, we would repeat the text with only slight change.

1. Prof. Xu Yi Da, the founder, and a Professor at HKBU, holds an impressive record with more than twenty years of research in machine learning, particularly in computer vision. He has transcended the realm of theoretical research by developing a host of research prototypes and software solutions, many of which are nearly ready for commercialization, pending the support of startup funding. Prof. Xu's rich professional background is augmented by his five-year experience as a software developer at IBM Australia. His combined academic and industry experience plays a crucial role in guiding the technological advancements of our products. Additionally, during his affiliation with the University of Technology Sydney, Prof. Xu was instrumental in establishing a formidable Industry Data Science research team. This team, at its height, consisted of 30 dedicated members including scholars, postdoctoral researchers, PhD students, and skilled engineers. They collectively embarked on over 20 research contracts with various sectors such as finance, government, transport, utilities, retail, e-commerce, defence, and the legal industry, working with both government and private entities.

At TadReamk, Prof. Xu is entrusted with the vital task of setting our strategic direction and garnering the resources essential for our company's growth and innovation.

2. Dr. Nguyen, serving as the Chief Research Officer, possesses an extensive wealth of expertise in the realm of image processing. His capabilities extend far beyond academic research; he excels at applying theoretical knowledge to practical solutions, a skill that is unmatched. He is responsible for defining the research trajectory across all facets of our work. His primary responsibilities include spearheading research initiatives and outlining strategic approaches to convert these research findings into tangible, real-world applications.

His team has been supported by two outstanding researchers:

- i. Mr. Chen Chun Hang, our current part-time researcher, is a final-year undergraduate student at HKUST. Despite his young age, he has excelled and emerged as the top candidate among all four summer interns (three of whom were hired as part of the STEM Internship program).

Chun Hang is a prime example of someone who has benefited from the STEM intern program. He joined TadReamk Limited as a summer STEM intern and later advanced to a part-time Research Associate role, contributing to TadReamk Create. He then progressed to become a Researcher, leading several other interns and junior staff. Upon graduating

from HKUST, he will join TadReamk Limited as a full-time employee. We also plan to apply for RTH for him in Aug, 2025.

He is also taking on the role of part-time business development. Despite his young age and relative inexperience in the business world, he has begun to excel in assisting with various business development tasks.

Previously, we had Dr. Wang Ningxia, who recently earned a PhD in Computer Science from HKBU and was instrumental in developing advanced detection models for Tadreamk Eye. Unfortunately, we had to let her go due to the delayed approval of the Talent Hub program, which caused significant cash flow issues.

- ii. Mr. Niu Shui, a co-founder and soon-to-graduate PhD student from HKBU, with the possibility to join the company full-time upon completion of his studies. Currently, he contributes as a research assistant, engaging in a range of initiatives related to image processing and machine learning.
- 3. Mr SUN, Xiaoyuan, whom is the chief Technology Officer, an experienced app developer with seven years of industry expertise, possesses a skill set that is highly sought after in Hong Kong. He serves as the lead developer, driving the future development of the company.

The team's efforts are also supported by external contractors and interns.

- 4. Mrs. Linda Chan, a former HR manager with vast corporate experience, has been an invaluable asset to our business. Her contributions extend far beyond the realm of human resources, touching nearly every facet of our operations, including external communications with various government entities and business stakeholders. As an accomplished business developer, Linda brings a wealth of experience in business growth strategies along with a profound knowledge of HR intricacies. Her expertise is particularly crucial for our startup, as she ensures fairness and efficiency in our talent acquisition and management practices, which are essential for our success.
- 5. Mr. Sun Cheng, the highest-ranked Master of Philosophy graduate from HKBU, has been an asset to her team. Despite the field of design philosophy being a departure from his prior academic pursuits, he has embraced the role with full commitment. His dedication and rigorous research methodologies have been seamlessly applied, contributing significantly to the team's efforts.

He departed from the company at the end of 2024 to pursue a different career path.

- 6. The company also benefited of having other outstanding long term interns, including:

- i. Ms. Carrie Lin – Has contributed significantly to several HR projects, including the STEM internship and YSIP recruitment.
- ii. Mr. Jason Gong – An undergraduate long-term research intern who will continue to support the company in technological endeavors for the foreseeable future.

11. Social and/or community impacts of the technology start-up's R&D work

Please note that although we have made significant progress over the past six months, our original goals and determination remain as strong as ever. Therefore, we have reused the same text from the 2023-2024 final report for this section:

Impact for small to medium enterprise (SME):

Our startup was founded with the mission of equipping small and medium-sized enterprises (SMEs) with an easy and affordable access to the entire spectrum of trademark technology. This encompasses everything from the initial registration process to the proactive mitigation of infringement risks.

Our commitment is to level the playing field, enabling SME businesses to secure their intellectual property with low cost and efficiency. We applied Artificial Intelligence that streamlines the registration of intellectual property, ensuring that creative individuals and businesses can protect their unique designs effectively. In addition, our service includes providing detailed descriptions and narratives for logo designs. This feature not only empowers clients with a clear understanding and artistic interpretation of their logos' strengths but also delivers this insight in an interpretable manner, bolstering their confidence in their brand identity.

Sharing an entrepreneur experience to the world

Prof. Xu's continuous commitment over the last ten years has borne fruit in the form of a large community of tens of thousands of machine learning followers across multiple social media channels. His dedication to fostering a sense of community within the machine learning sphere has been a key factor in this success. By actively engaging with his audience, Prof. Xu gives insightful updates and shares the personal journey of TadReamk, both the challenges and excitement of a startup. His narrative serves as inspiration for more talented people to embark on their entrepreneurs ventures.

During several trips to home (Australia), Prof. Xu always seize the opportunity to give an exciting presentation about his venture to the local start-up ecosystem. This initiative not only elevated TadReamk's profile but also highlighted Hong Kong's emerging status as a dynamic global center for technology start-ups. The strategic promotion of TadReamk, spearheaded by Prof. Xu, has indirectly played a role in enhancing Hong Kong's image as a vibrant cradle for technological innovation. Additionally, these activities have positively influenced the prestige of the Hong Kong Science and Technology Parks (HKSTP), further entrenching its standing in the technological landscape.

Media Coverage

The company's products and its founder were featured in the Takungpao newspaper on April 27, 2025. Tadreamk Limited has also been highlighted on various social media platforms, including the HKBU KTO Channels.

Exhibitions

Since the launch of our products, we have showcased them at various venues to attract potential customers and build publicity:

1. Guangdong-Hong Kong-Macao Greater Bay Area Intellectual Property Trade Expo (GBAIP) - November 2024
2. Business of IP Asia Forum (BIP Asia) - November 2024
3. InnoEx 2025